INCH-POUND
MIL-DTL-49285/14A
31 March 2010
SUPERSEDING
MIL-C-49285/14
19 October 1989

DETAIL SPECIFICATION SHEET

# CABLE, SPECIAL PURPOSE, ELECTRICAL, ONE INDIVIDUALLY SHIELDED PAIR OF 20 (7 X 28) AWG

Inactive for new design after 16 June 1997.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-49285.

Engineering information: See table I.

Note: Not for aerospace use.

TABLE I. Description, electrical.

| Electrical characteristics | Requirements |
|---|---|
| Nominal capacitance between conductors | 27 pF/ft (picofarads/foot) |
| Nominal capacitance between the conductor and the shield with one conductor tied to the shield | 49 pF/ft |
| Continuous working voltage | 300 V rms maximum between conductors |
| Shield dc resistance | 9.3 ohms/1,000 feet maximum |



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TABLE II. Description, physical.

| Components | Construction details |
|---|---|
| Number of pairs | One individually shielded |
| Conductor type and wire size | Stranded tinned copper 20 AWG |
| Conductor stranding | 7 X 28 AWG |
| Conductor insulation | Polypropylene |
| Conductor insulation thickness | 0.016 inch nominal thickness |
| Drain wire type and size | Stranded tinned copper 20 AWG |
| Drain wire stranding | 7 X 28 AWG |
| Jacket material | Polyvinyl chloride |
| Jacket thickness | 0.026 inch |
| Finished cable diameter | 0.214 inch maximum outside diameter |
| Cable style (UL) | 2092 |
| Tensile strength (jacket) | 2,000 pounds per square inch minimum |
| Elongation (jacket) | 150 percent minimum |
| Overall cabling lay length | 8.0 twists per foot ± 10 percent |

REQUIREMENTS:

Design and construction: (see table II):

Shield color code. The shield color shall be blue.

Shield location and orientation. The polyester aluminum shield is to be located on the outer circumference of the pair, with the aluminum foil side outward. Insulation on the interior of the shield is to be complete with no aluminum available for contact.

Drain wire location. The drain wire is to be spirally located between the shield and jacket, and is to be in continuous contact with the aluminum surface of the foil shield throughout the cable.

Cable temperature rating. The cable temperature rating shall be -20°C to +80°C.

Conductor color coding. The first conductor is black, the second conductor is natural (clear).

Shield integrity test.*

| Frequency | Response ** |
|---|---|
| 100 kHz | 115 dB below reference |
| 500 kHz | 106 dB below reference |
| 1 MHz | 98 dB below reference |
| 5 MHz | 80 dB below reference |



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| | |
|---|---|
| 10 MHz | 71 dB below reference |
| 15 MHz | 66 dB below reference |
| 20 MHz | 61 dB below reference |
| 25 MHz | 57 dB below reference |
| 30 MHz | 54 dB below reference |

* Any two readings may deviate to a value of 0.9 times the stated limits.

** All values are a minimum numerical value.

Fixture resonance will occur between 40 and 70 MHz.

At 100 MHz: At least 41 dB below reference.

Flammability. The cable shall pass UL 2556 flame test requirements.

Crosstalk test limits. Crosstalk testing is not required for this configuration.

Durometer hardness. The cable jacket shall have a “Shore A” hardness of 82 ± 5.

Part or Identifying Number (PIN): The PIN shall be M49285/14.

Changes from previous issue. The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Referenced documents. This document references MIL-DTL-49285 and UL 2556.

CONCLUDING MATERIAL

Custodians:
Army – CR
Navy – SH
Air Force – 85
DLA - CC

Preparing activity:
DLA - CC

Review activities:
Army – MI
Air Force – 99
DLA – IS

(Project 6145-2009-018)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change. You should verify the currency of the information above using the ASSIST Online database at https://assist.daps.dla.mil.



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